

**HOW TO USE THIS SET**

20 questions, mixed difficulty, 4 case-based. Attempt every question before checking the answer key on the last page.

4 Easy

10 Medium

6 Hard

1.**EASY**

The pancreas is best classified as:

- A. A lymphoid organ
- B. A purely ductal structure with no secretory acini
- C. A purely exocrine gland
- D. A mixed exocrine and endocrine gland
- E. A purely endocrine gland

2.**EASY**

Which cell type forms the acinus and secretes digestive enzymes under CCK control?

- A. Islet alpha cell
- B. Acinar cell
- C. Ductal columnar cell
- D. Beta cell
- E. Centroacinar cell

3.**EASY**

Which islet cell type is most numerous, located centrally within the islet, and secretes insulin?

- A. Delta cell
- B. Alpha cell
- C. Acinar cell
- D. F (PP) cell
- E. Beta cell

4.**EASY**

Which duct segment is the smallest, beginning directly from centroacinar cells inside the acinus?

- A. Main pancreatic duct
- B. Intralobular duct
- C. Common bile duct
- D. Interlobular duct
- E. Intercalated duct

5.**MEDIUM****CASE**

A histology slide shows several pyramidal cells forming an acinus, with a basophilic base rich in RER and eosinophilic apical granules, secreting enzymes in response to CCK.

Which cell type is this, and what does it secrete?

- A. Centroacinar cell; bicarbonate-rich fluid
- B. Beta cell; insulin
- C. Delta cell; somatostatin
- D. Acinar cell; amylase, lipases and proteases
- E. Ductal cell; mucus only

6.

MEDIUM

Centroacinar cells are distinguished from acinar cells by which feature?

- A. Centroacinar cells are the most numerous cell in the islet
- B. Centroacinar cells are found only in the duct system outside the acinus
- C. Centroacinar cells sit inside the acinus lumen at the origin of the intercalated duct and secrete watery bicarbonate-rich fluid under secretin control
- D. Centroacinar cells secrete insulin
- E. Centroacinar cells have eosinophilic apical zymogen granules

7.

MEDIUM

Which duct segment is formed by multiple intercalated ducts joining within one lobule, and lined by low columnar epithelium?

- A. Intralobular duct
- B. Interlobular duct
- C. Intercalated duct
- D. Common bile duct
- E. Main pancreatic duct

8.

MEDIUM

Where in the pancreas are islets of Langerhans most numerous?

- A. Head
- B. Body only
- C. Uncinate process
- D. Evenly distributed with no regional preference
- E. Tail

9.

MEDIUM

Which islet cell secretes somatostatin, inhibiting both glucagon and insulin release?

- A. Acinar cell
- B. F (PP) cell
- C. Beta cell
- D. Delta cell
- E. Alpha cell

10.

MEDIUM

Which structure is formed by the convergence of many interlobular ducts, running from the tail to the head of the pancreas, and opening at the major duodenal papilla?

- A. Accessory pancreatic duct only
- B. Intralobular duct
- C. Intercalated duct
- D. Main pancreatic duct
- E. Bile canaliculus

11.

MEDIUM

F (PP) cells of the pancreatic islet secrete pancreatic polypeptide, which functions to:

- A. Inhibit pancreatic exocrine secretion and relax the gallbladder while decreasing bile secretion
- B. Increase blood glucose
- C. Increase gastric acid secretion
- D. Stimulate glucagon release
- E. Directly stimulate insulin release

12.

MEDIUM

Interlobular ducts, found in the septa between lobules, differ from intralobular ducts by:

- A. Never converging into the main pancreatic duct
- B. Being columnar and potentially containing goblet cells, and being surrounded by abundant connective tissue
- C. Secreting insulin directly into the ducts
- D. Being lined by simple squamous epithelium only
- E. Beginning directly at the centroacinar cell

13.

MEDIUM

What is the ordered sequence of the exocrine pancreatic duct system, from acinus to duodenum?

- A. Acinus → common bile duct → duodenum directly
- B. Acinus → interlobular duct → intralobular duct → intercalated duct → main pancreatic duct → duodenum
- C. Acinus → intralobular duct → main pancreatic duct → duodenum
- D. Acinus → main pancreatic duct → intercalated duct → duodenum
- E. Acinus → intercalated duct → intralobular duct → interlobular duct → main pancreatic duct → duodenum

14.

MEDIUM

Which statement correctly compares alpha and beta cells of the islet by position and proportion?

- A. Both are found only at the islet periphery in equal proportion
- B. Beta cells are peripheral and secrete glucagon
- C. Alpha and beta cells cannot be distinguished by position
- D. Beta cells are central (~70%, insulin); alpha cells are peripheral (~20%, glucagon)
- E. Alpha cells are central (~70%); beta cells are peripheral (~20%)

15.

HARD

CASE

A patient with chronic pancreatitis develops progressive destruction of pancreatic acini with relative sparing of the islets of Langerhans on biopsy.

Based on the distinct secretory roles described in this lecture, which functional deficit would be expected first?

- A. Loss of glucagon secretion specifically
- B. No functional deficit, since acini are not essential
- C. Diabetes mellitus from loss of insulin secretion
- D. Loss of somatostatin secretion causing unopposed insulin release
- E. Malabsorption from loss of digestive enzymes (amylase, lipases, proteases) despite preserved hormone secretion

16.

HARD

A tumour arising from centroacinar-lineage cells would be expected to disrupt secretion of which specific pancreatic product, given the described control mechanism?

- A. Glucagon, since centroacinar cells are equivalent to alpha cells
- B. Pancreatic polypeptide, since centroacinar cells are equivalent to F cells
- C. Insulin, since centroacinar cells are a type of beta cell
- D. Bicarbonate-rich fluid, since centroacinar cells secrete this under secretin control at the start of the duct system
- E. Gastrin, since centroacinar cells are enteroendocrine in origin

17.

HARD

CASE

A researcher blocks CCK receptors selectively on pancreatic acinar cells in an animal model, while secretin signalling to centroacinar cells remains intact.

Which secretory response would be most impaired?

- A. Insulin release from beta cells
- B. Release of digestive enzymes (amylase, lipases, proteases) from acinar cells in response to a fatty/protein meal
- C. Glucagon release from alpha cells
- D. Bicarbonate-rich fluid secretion in response to a duodenal acid load
- E. Somatostatin release from delta cells

18.

HARD

On an unlabelled histology field, small clusters of pale, polygonal cells separated by a fenestrated capillary network and surrounded by a reticular connective tissue capsule are seen scattered among darker, more granular acinar tissue. What is the best identification, and what is the significance of the capillary network?

- A. Centroacinar cell clusters; the capillaries provide bicarbonate
- B. Interlobular ducts; the capillaries drain bile
- C. Intercalated ducts; the capillaries are incidental
- D. Lymphoid follicles; the capillaries carry lymph
- E. Islets of Langerhans; the fenestrated capillaries allow efficient release of hormones directly into the bloodstream

19.

HARD

A patient has a tumour secreting excess pancreatic polypeptide (a PPoma). Based on F/PP cell function, which of the following would be an expected consequence?

- A. Increased insulin secretion with hypoglycaemia
- B. Increased glucagon secretion with hyperglycaemia
- C. Reduced pancreatic exocrine secretion (enzymes and bicarbonate) and a relaxed, poorly contracting gallbladder
- D. No effect on exocrine or biliary function
- E. Increased pancreatic enzyme and bicarbonate secretion with gallbladder contraction

20.

HARD

CASE

A surgical specimen shows a duct lined by columnar epithelium containing goblet cells, surrounded by abundant connective tissue, located in the septa between lobules.

Which duct segment is this, and what does the goblet cell content suggest about its position in the duct hierarchy?

- A. Interlobular duct; goblet cells and abundant surrounding connective tissue are consistent with a larger, more distal duct rather than the small intercalated or intralobular segments
- B. Main pancreatic duct; it opens directly into the stomach
- C. Centroacinar segment; it has no true duct wall
- D. Intralobular duct; it lies deep within the lobule near the acinus
- E. Intercalated duct; it is the smallest, most proximal segment

ANSWER KEY & RATIONALE

Don't peek until you've committed to an answer on every question.

Q	Ans	Why
1	D	A mixed exocrine and endocrine gland
2	B	Acinar cell
3	E	Beta cell
4	E	Intercalated duct
5	D	Acinar cell; amylase, lipases and proteases
6	C	Centroacinar cells sit inside the acinus lumen at the origin of the intercalated duct and secrete watery bicarbonate-rich fluid under secretin control
7	A	Intralobular duct
8	E	Tail
9	D	Delta cell
10	D	Main pancreatic duct
11	A	Inhibit pancreatic exocrine secretion and relax the gallbladder while decreasing bile secretion
12	B	Being columnar and potentially containing goblet cells, and being surrounded by abundant connective tissue
13	E	Acinus → intercalated duct → intralobular duct → interlobular duct → main pancreatic duct → duodenum
14	D	Beta cells are central (~70%, insulin); alpha cells are peripheral (~20%, glucagon)
15	E	Malabsorption from loss of digestive enzymes (amylase, lipases, proteases) despite preserved hormone secretion
16	D	Bicarbonate-rich fluid, since centroacinar cells secrete this under secretin control at the start of the duct system
17	B	Release of digestive enzymes (amylase, lipases, proteases) from acinar cells in response to a fatty/protein meal
18	E	Islets of Langerhans; the fenestrated capillaries allow efficient release of hormones directly into the bloodstream
19	C	Reduced pancreatic exocrine secretion (enzymes and bicarbonate) and a relaxed, poorly contracting gallbladder
20	A	Interlobular duct; goblet cells and abundant surrounding connective tissue are consistent with a larger, more distal duct rather than the small intercalated or intralobular segments

Histology GIT Block · MEDucation by Mattin Majid · Source: GIT Histology LG 3, Dr Hero K. Mustafa

Recalled and source questions are treated as evidence of what gets asked, not as a source of correct answers — verify against your lecture notes.